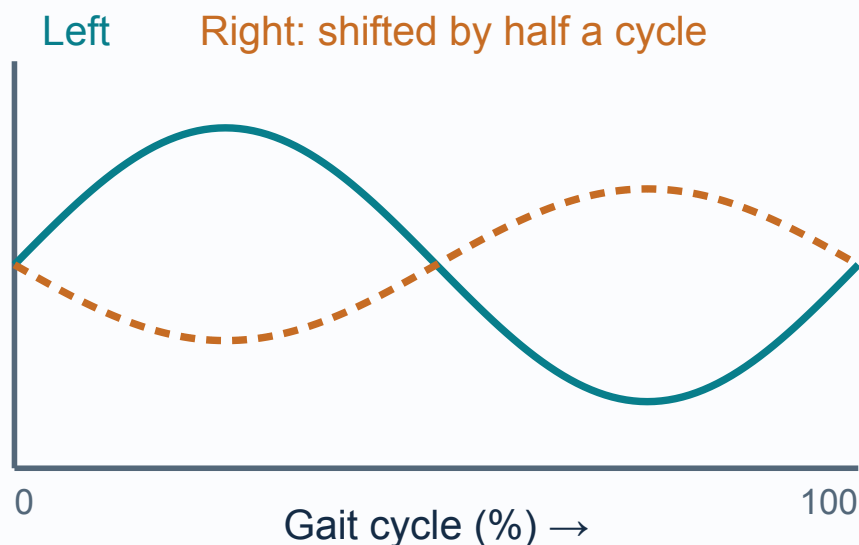


Geometry, symmetry, and learned representations of gait

How do we test whether a video model captures useful movement information?

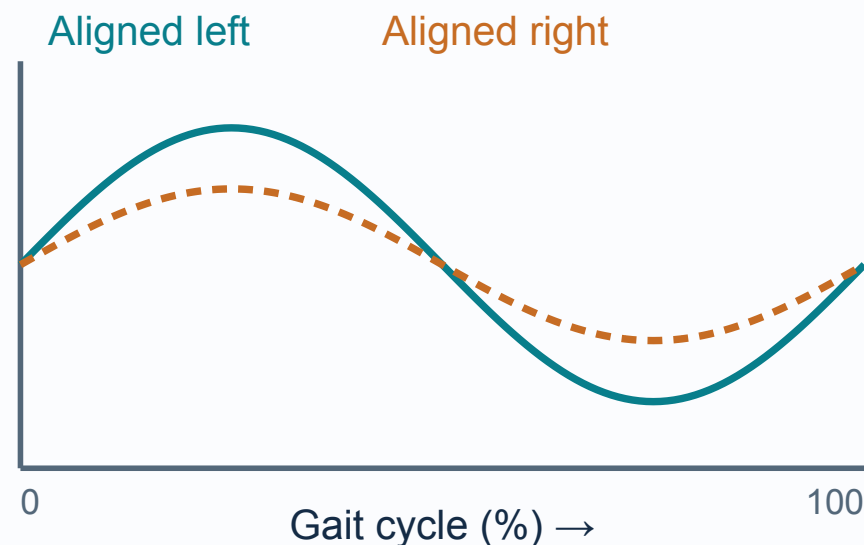
Compare corresponding phases



Angle (schematic units). Same unequal amplitudes in both panels; only the right-side phase is aligned.

Proposed analysis; these are synthetic curves, not measured participant results.

Then inspect the difference



Movement is a relationship between body parts

Angles describe posture

The bend of a knee depends on the thigh and shin together. Its projected angle survives translation and uniform image scaling.

Timing describes walking

Left and right legs usually alternate. Comparing them requires matching corresponding phases of the gait cycle.

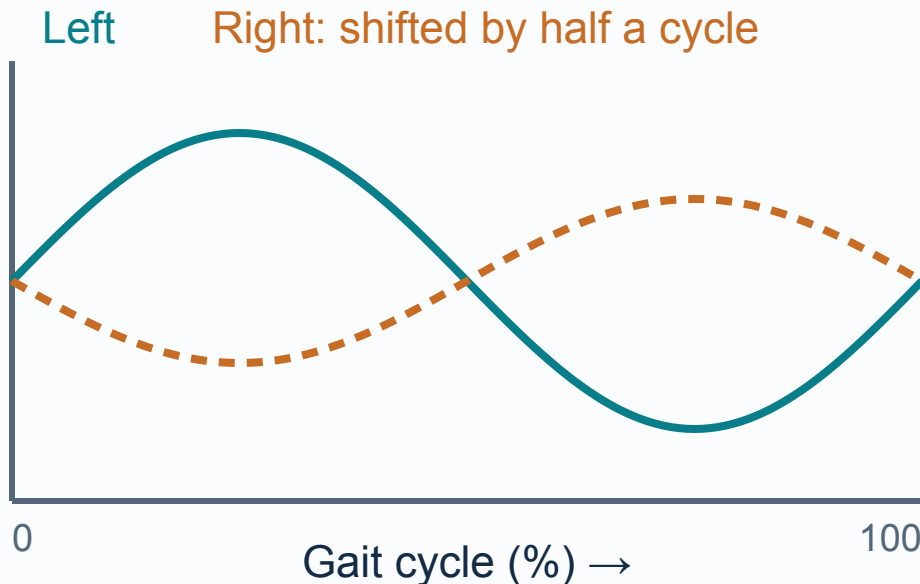
Health motivates the question

PD can involve unequal arm swing. MS can alter coordination between joints. The pattern varies between people.

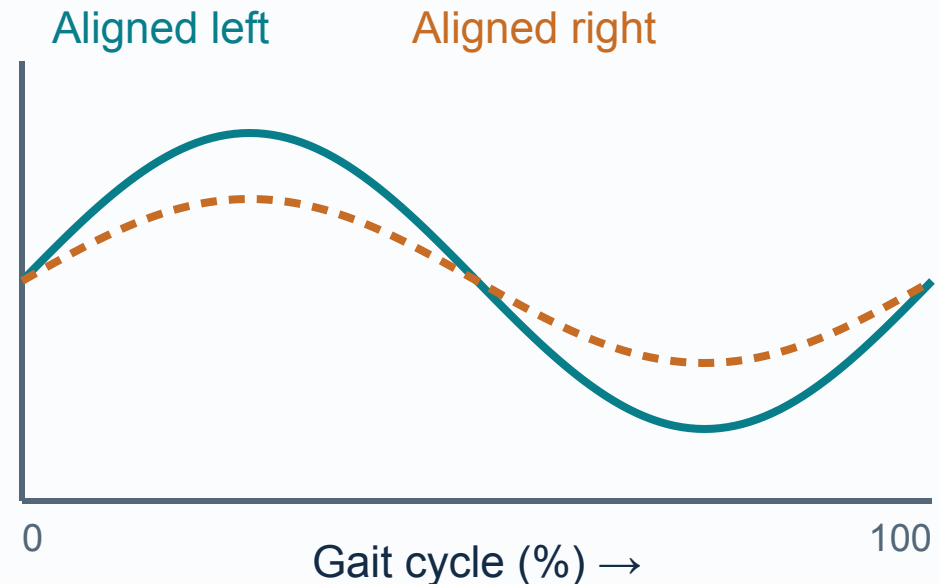
Clinical motivation comes from separate gait studies; these mechanisms have not been established in our video collection.

Compare the same movement phase on both sides

Compare corresponding phases



Then inspect the difference



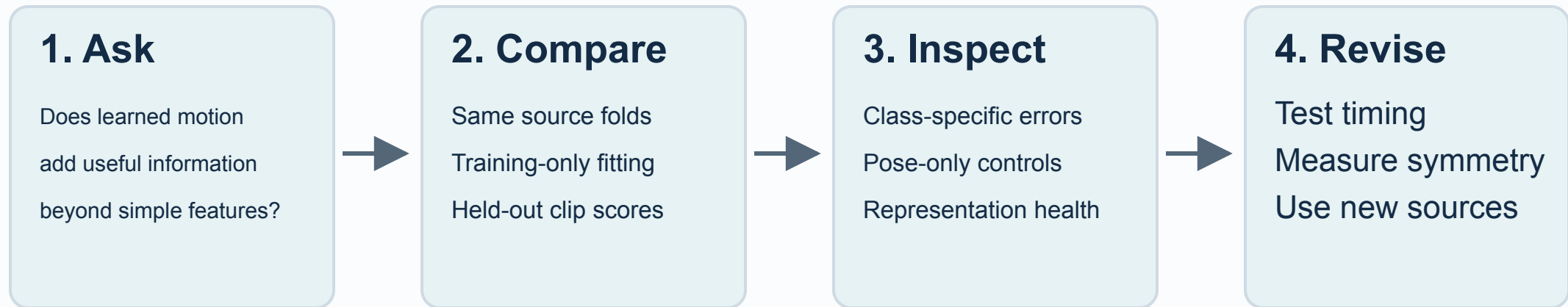
Angle (schematic units). Same unequal amplitudes in both panels; only the right-side phase is aligned.

Proposed analysis; these are synthetic curves, not measured participant results.

An unequal curve could reflect movement, viewpoint, tracking error, or alignment error. Each explanation needs a control.

Start with a comparison that can disappoint us

Working null: no added value from the learned representation in this comparison.



This organizing hypothesis was not a preregistered significance test; all retained estimates are developmental.

We also ask whether a predictive signal depends on motion and symmetry, beyond pose summaries or recording cues.

Follow the question from raw video to evaluation

00

Inventory

Clips and sources

01

Pose

Coordinates + visibility

02

Tokens

Local joint-time units

03

Train

800-update demo

04

Continue

400 more updates

05

Visualize

Retained run failed

06

Evaluate

Demo + frozen results

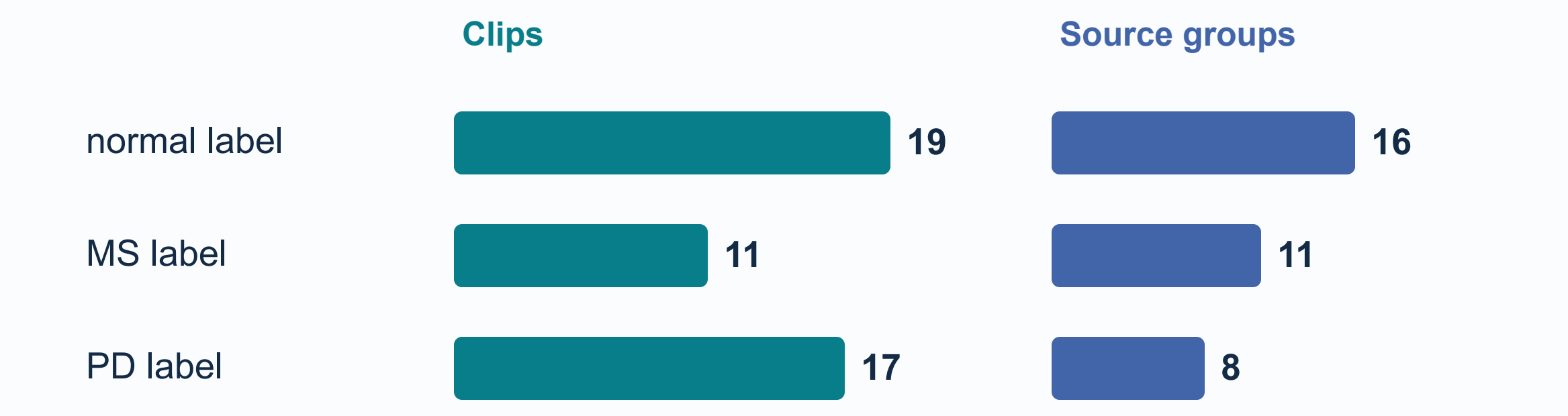
R1

Reference

5 folds × 1,000 updates

Notebook demonstrations and the frozen five-fold reference answer different questions. Their scores stay separate throughout.

Forty-seven clips do not imply forty-seven people

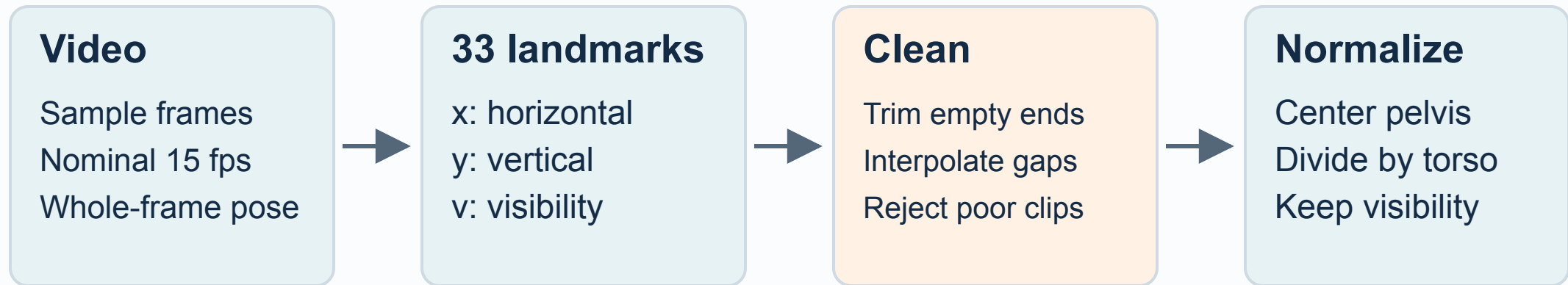


A source is an original video identifier. Different sources can still include the same person.

The collection has 49 input clips from 37 sources; 47 clips have usable cached poses. Source identifiers are provisional units for splitting.

Convert video frames into estimated 2D skeletons

Every retained frame has 33×3 values; the third channel is visibility, not depth.

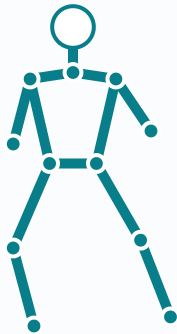


Cleaning can invent intermediate coordinates. The cache lacks explicit validity and timestamp channels.

The cache contains 8,693 frames and yields 481 model windows. Pose estimates remain subject to tracking and camera errors.

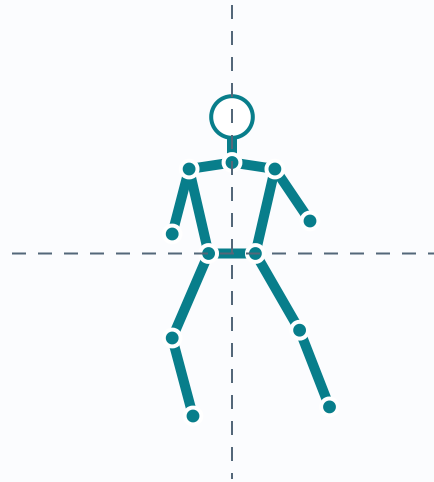
Normalize position and scale within each frame

Estimated image pose



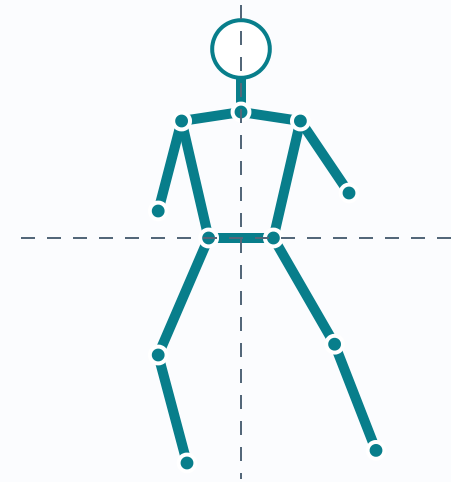
Coordinates q

Subtract pelvis p



$q - p$

Divide by torso s



$q' = (q - p) / s$

Schematic. Preserved within a frame: angles and distance ratios. Removed: root trajectory and absolute scale.

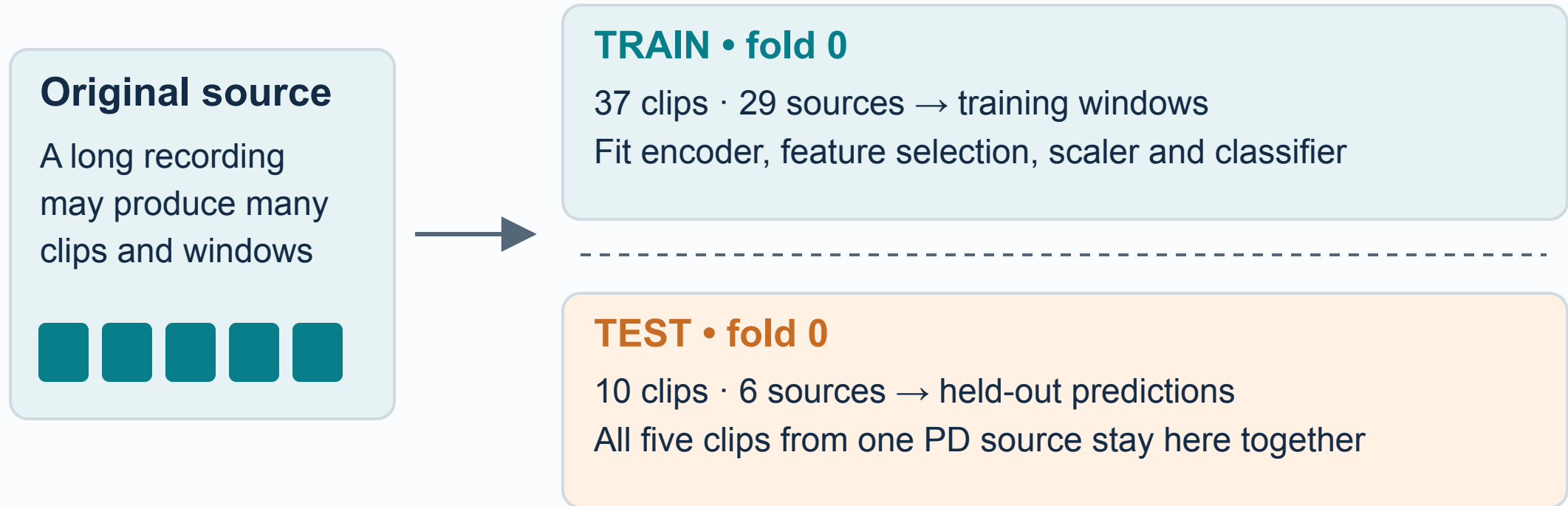
The same subtraction and scale apply to every joint in a frame, preserving its 2D angles and distance ratios.

Fixed tensor dimensions do not ensure physical fidelity

Operation	What remains	What changes
Reshape into tokens	Same ordered input values	Array layout; then learned projection
Center + torso scale	2D angles; distance ratios	Root motion; per-frame scale
Rotate / scale / reflect	2D shape up to similarity	Coordinates; signed laterality
Interpolate / pad	Fixed input dimensions	Unobserved time points are filled
Encode + pool	A learned 96-value vector	Detailed coordinates and order

We preserve a defined representation of the data. Exact preservation of the original dataset's geometry is not established.

Assign sources before fitting or sampling windows

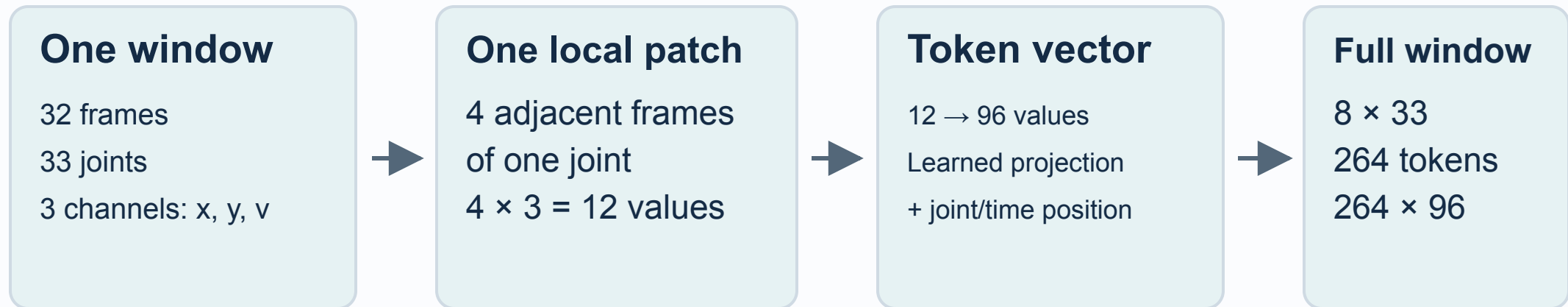


Grouping prevents this source overlap; participant identity and recording confounding remain unresolved.

All training, including label-free pretraining, excludes the held-out sources. Both model branches share this boundary.

Give the model small, ordered joint-time patches

Local order is explicit before compression into a learned representation.



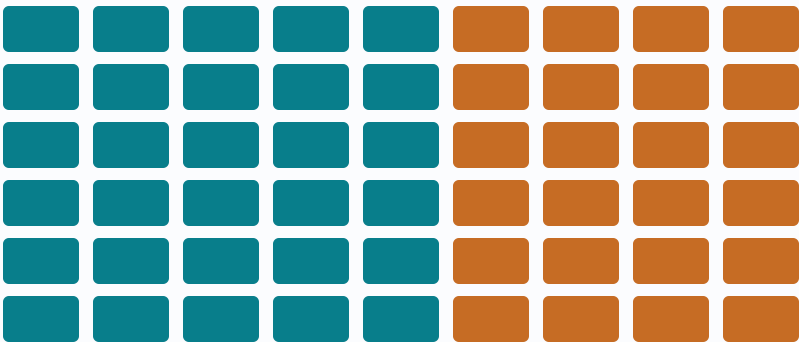
The 32-frame window is roughly 2 seconds at nominal 15 fps; actual cached timing is not fully verified.

A token represents four adjacent observations of one joint. Position vectors identify its joint and time block.

Let joints alternate between evidence and targets

Historical fixed mask

Joint groups →



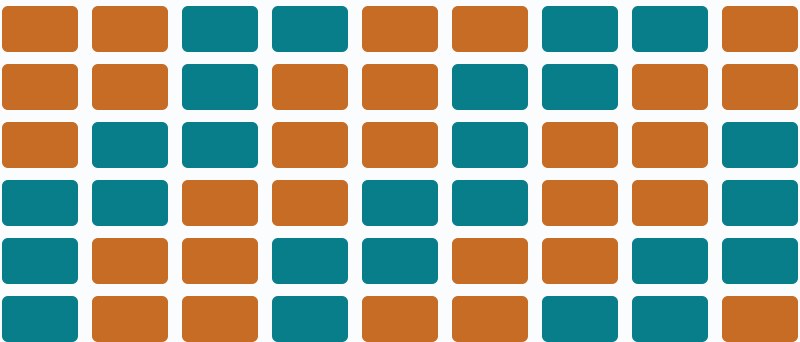
Some joint identities never enter context.

■ Visible context ■ Hidden prediction target

Schematic grids; actual layout is 8 time blocks × 33 joints. Repaired target fraction is about 60%.

Current varying masks

Joint groups →



Each joint can serve as context and target.

The repaired sampler hides varying anatomical regions over time. The earlier fixed mask withheld the same twelve joints on every update.

Tell the predictor which hidden position to fill

A target needs a location in the skeleton and in time.

Historical hidden slots

mask

mask

mask

Repaired hidden slots

mask + L knee, t_1

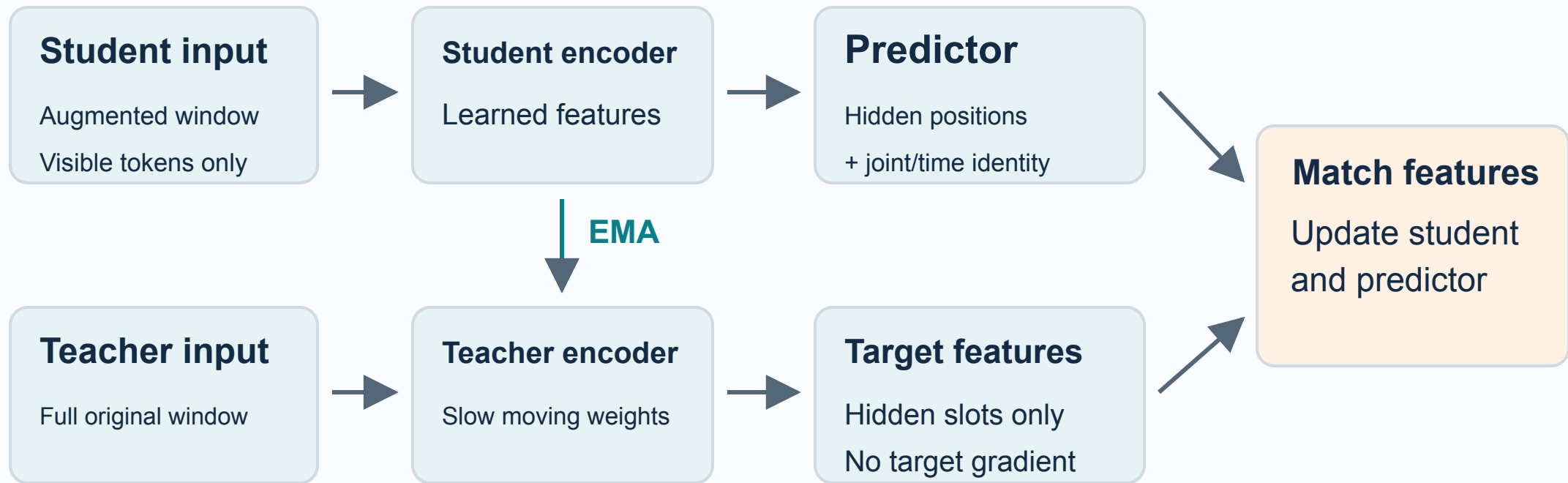
mask + R ankle, t_2

mask + hip, t_3

Position vectors distinguish the prediction requests; this structural repair does not establish a score gain.

A shared mask token needs joint and time information, so different missing positions can request different predictions.

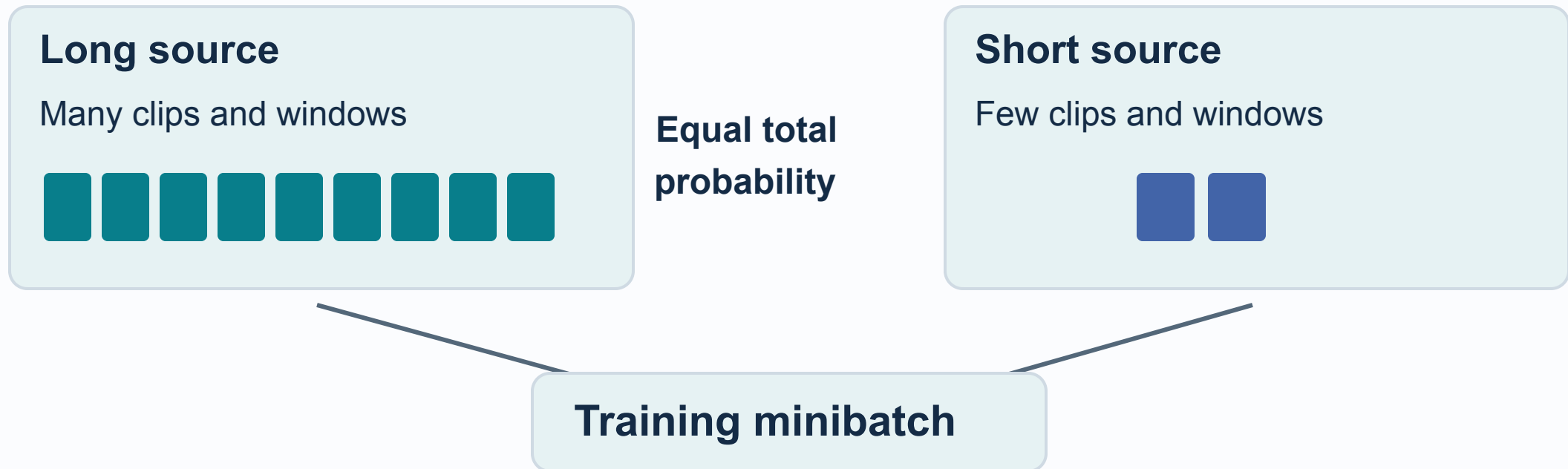
Predict a hidden part of the teacher's representation



EMA = exponential moving average. Centering and sharpening shape the loss; collapse is checked separately.

JEPA matches feature vectors. Its targets come from a slowly updated teacher, rather than clinical labels.

Keep a long recording from dominating the updates

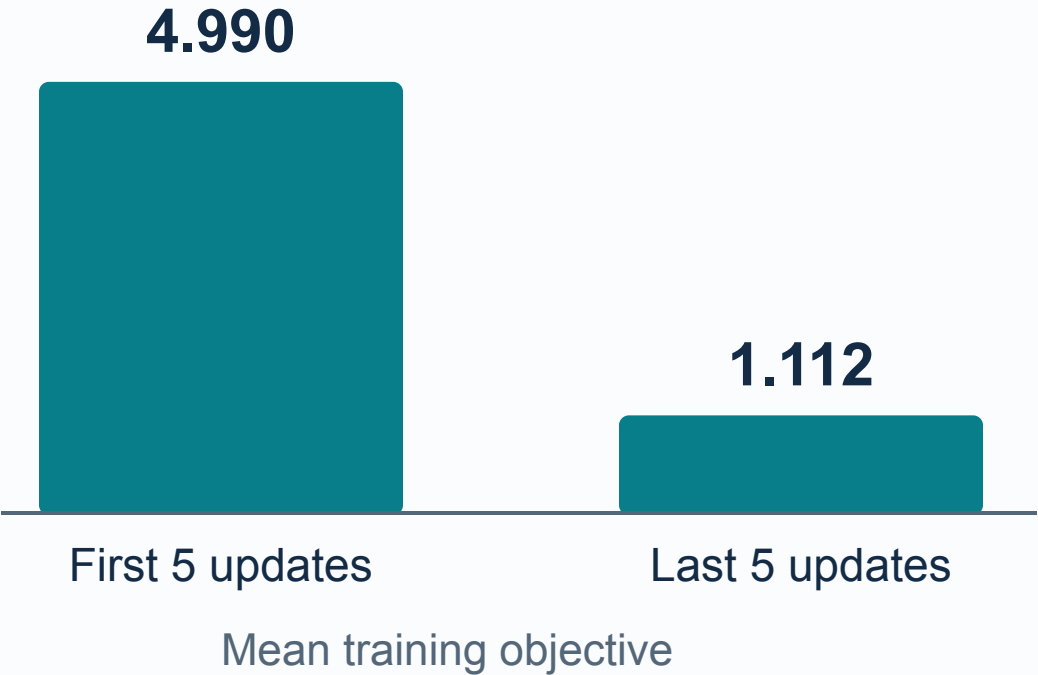


The current SSL loop does not read condition labels. Labels enter later when fitting the classifier.

Each training source receives equal total sampling probability. The selected windows retain their defined dimensions.

Loss falls while the representation retains variation

Notebook 03 • one training fold • 800 updates



Effective rank: 9.5

Feature spread within a batch
Batch size 32 → ceiling 31
A training diagnostic;
no clinical meaning assigned.

This retained 800-update demonstration uses 395 windows from 29 training sources. Loss reduction alone does not establish useful gait features.

The held-out probe stays at the same score



The retained continuation changed the feature spread (final rank 10.4), with no probe-score gain.

This is a single-fold demonstration with restarted training state, not a five-fold or continuous 1,200-step run.

The before/after result separates optimization activity from improvement on the selected evaluation task.

Class separation can enter through the training objective

Historical progressive route

Normal-only pretraining was followed by MS/PD stages and a class-aware VICReg extension. Condition labels influenced the representation objective.

Current notebook route

All eligible training sources enter label-free JEPA training. Labels fit a separate classifier to frozen clip embeddings.

A claim that earlier clusters emerged without label information would overstate what that training procedure tested.

The retained visualization run stopped before its results

Observed execution

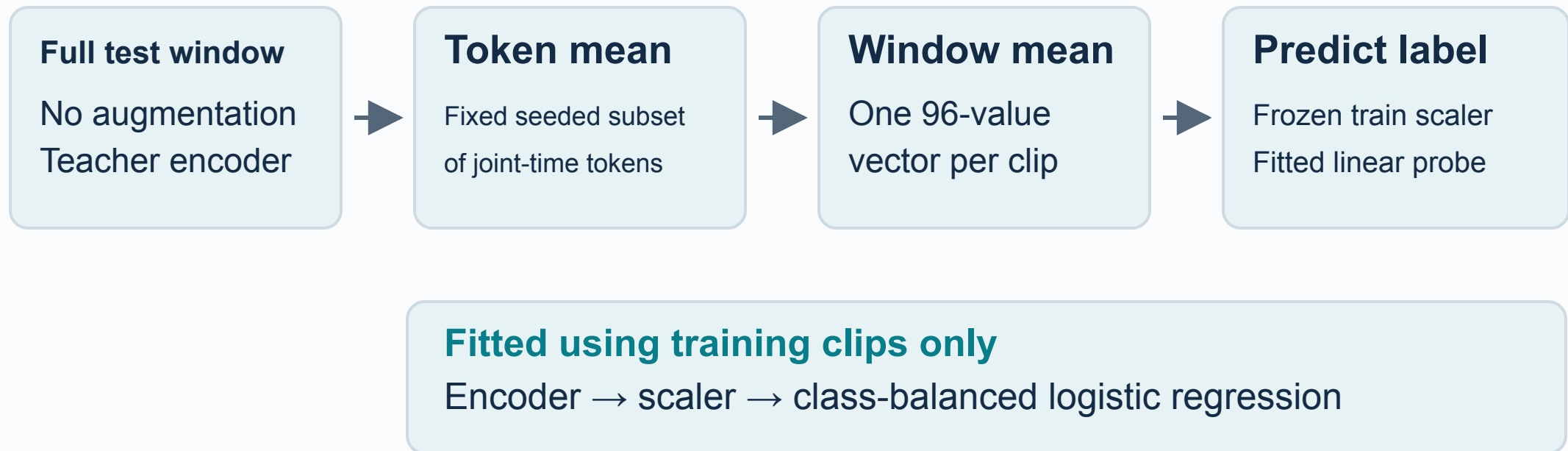
A loading cell raised `FileNotFoundError`. The intended embedding plots and downstream metrics were not executed.

What can be concluded

The saved run supplies no current cluster evidence. Checkpoints now exist, but the notebook still needs a successful rerun.

A two-dimensional embedding plot would remain exploratory after a successful execution; held-out prediction and explicit symmetry tests are still needed.

Freeze the encoder, then fit a small classifier



Averaging reduces the representation. Retention of phase timing and left–right differences has not been tested.

A linear probe is a classifier fitted to fixed features. It measures whether those features support the chosen labels.

Compare learned vectors with simple angle summaries



No explicit cycle-phase alignment is used in these summary features.

Implementation limitation: the left ankle range slot copies the right ankle range; the saved scores are unchanged.

Both branches use the same clips and source folds. The baseline summarizes angles; JEPA uses joint coordinates and visibility.

Keep training budget and test coverage beside the score

03 → 04 demonstration

Fold 0 only
800 + 400 from weights
Probe F1: 0.600 → 0.600

06 demonstration

Fold 0 only; fresh model
500 updates
S-JEPA 0.644 · RF 0.915

Frozen older reference

All five source folds
1,000 updates per fold
S-JEPA 0.438 · RF 0.667
Before final centering fix

All displayed scores are macro F1, but their training budgets and evaluation coverage differ.

The larger fold-0 scores are not an updated five-fold estimate or evidence that a shorter training run is better.

The current corrected trainer has notebook demonstrations. Its final centering repair has no retained five-fold rerun.

Give each label equal weight in macro F1

Accuracy

Count correct clip predictions and divide by all clips. A source with several clips contributes several decisions.

Per-class F1

Combine precision and recall. Precision asks how often a label's predictions are right; recall asks how many of its clips are found.

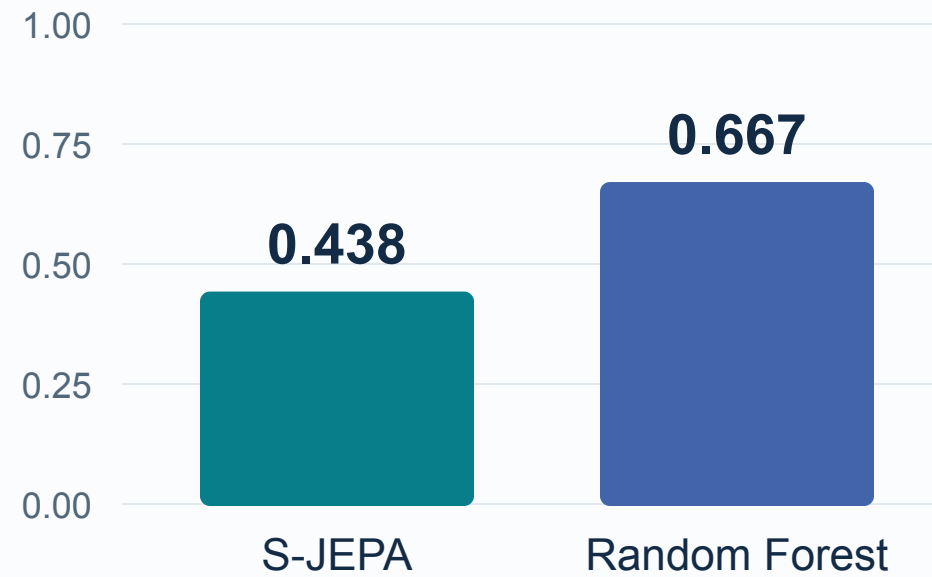
Macro F1

Average the three class F1 scores equally. A large class cannot dominate this average through its clip count alone.

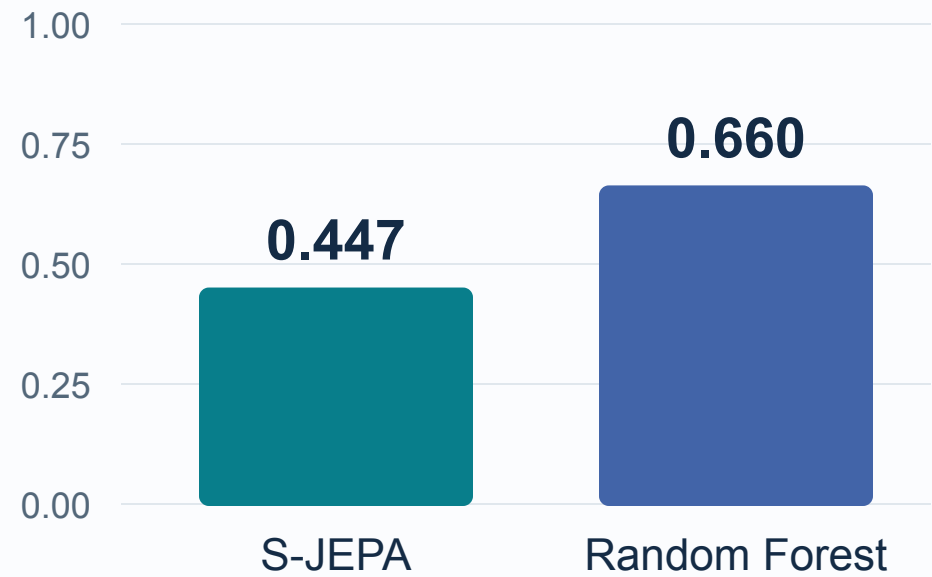
Pooled results combine one held-out prediction per clip. A mean across fold scores is a different summary; fold spread is not a confidence interval.

The retained learned representation trails the angle baseline

Macro F1



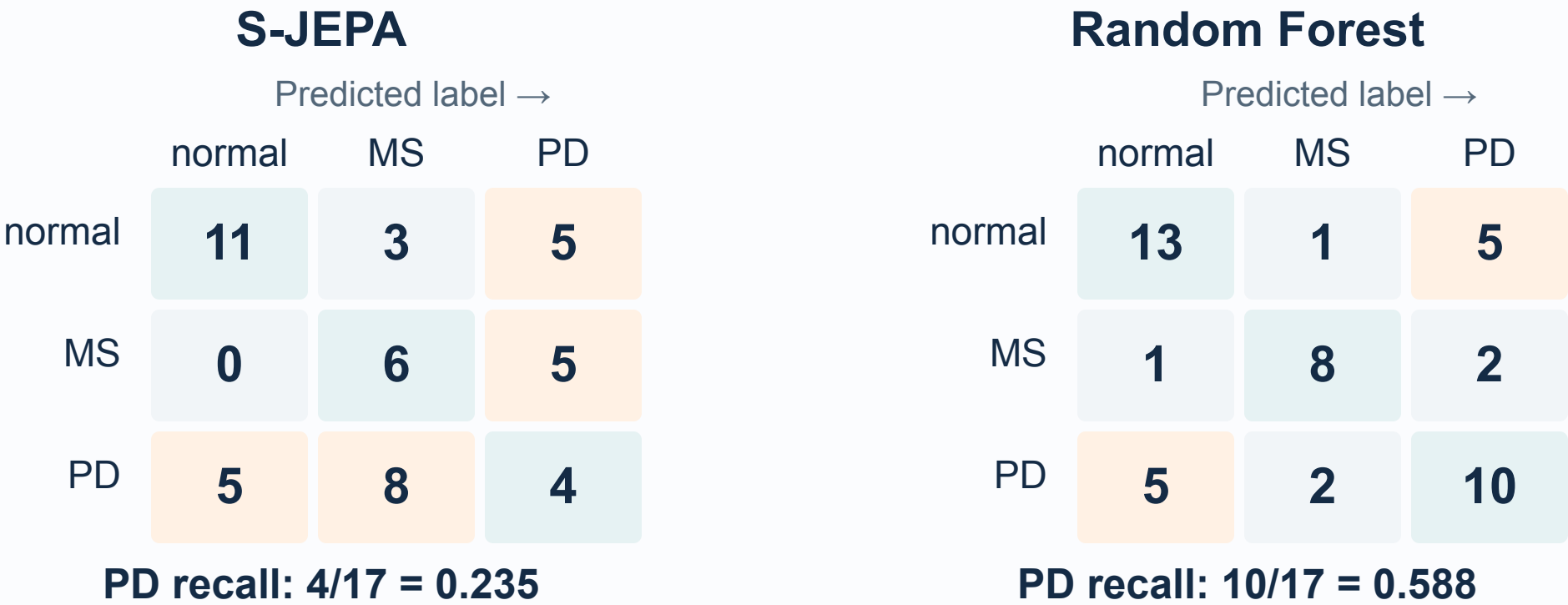
Accuracy



Frozen reference before final centering fix · 47 clips · 35 sources · one seed · no confidence intervals

The descriptive macro-F1 difference is -0.228 for S-JEPA relative to Random Forest. No group-aware confidence interval has been retained.

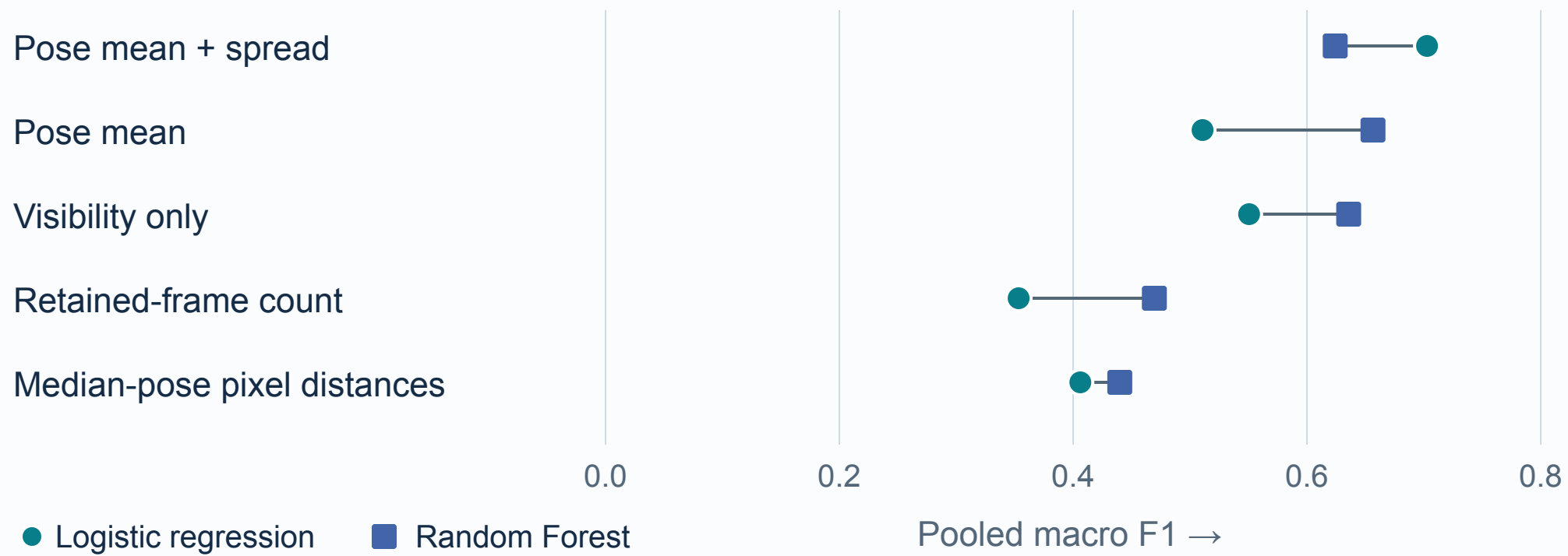
The PD-labelled clips expose a large weakness



Rows are true labels. S-JEPA sends 8 PD clips to MS; the Random Forest sends 5 PD clips to normal.

S-JEPA correctly labels 4 of 17 PD clips; Random Forest correctly labels 10. These clips come from eight PD sources.

Strong predictions survive without ordered motion



Pose mean and spread reach 0.703 with logistic regression. Visibility alone reaches 0.636 with RF, warranting closer acquisition controls.

Label prediction leaves the mechanism unresolved

Observed

Averaged pose features and visibility summaries predict many held-out clip labels in this source-grouped collection.

Plausible explanations

Posture, movement amplitude, camera view, tracking quality and source selection may each contribute to the scores.

Needed comparison

Match acquisition conditions and remove a specific type of information, such as frame order, before attributing its contribution.

The strongest saved simple control exceeds the angle RF. Describing the angle baseline as strongest overall would be inaccurate.

Grouping improves evaluation, but uncertainty remains

Training exposure

Source-uniform sampling
Limits domination by
long source recordings

Reported scores

One prediction per clip
47 clips in total
Sources have unequal
numbers of clips

Future inference

Verify participants
Bootstrap source groups
or participant groups
Use a fresh test cohort

Five-fold grouping protects the source boundary.

A confidence interval requires an explicit uncertainty calculation.

Current estimates reuse an inspected development collection, so they cannot support confirmatory clinical claims.

Source grouping addresses recording overlap. Verified participant identities and an untouched cohort would support stronger generalization claims.

WHAT THE EXPERIMENT HAS ESTABLISHED

Separate implementation checks from physical claims

Question	Retained evidence	Interpretation
Can repaired training run?	Loss falls; batch features vary	The objective can be optimized
Did further fold-0 training help?	Macro F1 stays 0.600	No observed probe gain
Does frozen S-JEPA beat angle RF?	0.438 versus 0.667	Reference favors angle RF
Is the signal uniquely temporal?	Pose summaries reach 0.703	This control needs no frame order
Has symmetry been measured?	No aligned bilateral or reflection result	Physical mechanism remains open

Measure reflection, temporal order and aligned asymmetry

Reflection test

Mirror x and relabel joints
Compare model features
under the defined mapping



Time-order control

Keep each pose intact
Shuffle frame order
Measure the score change



Symmetry measure

Align gait-cycle phase
Compare left/right angles
Control camera viewpoint



Proposed experiments: no measured equivariance, temporal ablation or phase-aligned asymmetry result is available yet.

Schematic. Predefine the comparison and fit only on training sources before measuring the held-out result.

Equivariance means an input transformation produces a defined corresponding output transformation. The network has no demonstrated guarantee of it.

Resolve input quality before expanding the claims

Each next experiment resolves a specific ambiguity in the current evidence.

1. Verify the input

- Review pose overlays
- Keep time / validity masks
- Check ankle-angle features
- Complete notebook 05

2. Test the mechanism

- Temporal-shuffle control
- Reflection comparison
- Phase-aligned asymmetry
- Matched acquisition cues

3. Strengthen inference

- Verify participant IDs
- Lock model decisions
- Evaluate fresh sources
- Report grouped uncertainty

The present study supports an inspectable development workflow; additional data and tests are still needed.

A corrected five-fold rerun should identify the final centering logic and corrected feature construction before comparison with the frozen reference.

Start with retained outputs, then run the intended protocol

```
# From experiments/multiple-sclerosis
uv run jupyter lab

# Rebuild the SVGs from saved evidence
python3 scripts/scripts_make_progress_diagrams.py
```

Notebook sequence

00 → 01 → 02 establishes inputs.
03 → 04 trains and probes.
05 needs successful execution.
06 separates demo and reference.

Saved reference

artifacts/runs/r1_g1_1k_s42/
Read results, out-of-fold predictions and the
run manifest together.

EVIDENCE AND READING

A developing physical model needs measured physical tests

The workflow connects articulated pose to a trainable JEPA representation and a source-grouped evaluation. Retained comparisons expose limits in predictive performance and leave the contribution of symmetry unresolved.

Study evidence: `docs/08-0913-PROGRESS.MD` ; notebooks 00-06; `artifacts/runs/r1_g1_1k_s42/results.json` ; `artifacts/eval/g1/E0_results.json` . Detailed provenance appears in speaker notes.

Clinical motivation: Lewek et al., arm swing in early Parkinson's disease; Pau et al., lower-limb coordination in multiple sclerosis.

Methods: Skeleton-based JEPA, ECCV 2024; VICReg.